DS	**Session**	Hashing	**Question Information**	Level 1 Challenge 91

Problem Description

Everyone knows that some Pikachus despise becoming Raichus. (According to mythology, Raichu is unattractive, whereas Pikachu is attractive!)

How do we track down these unique Pikachus who despise evolution? Because you're friends with the insane Poke'mon trainer Ash Catch'Em, he devised a random method that is absolutely incorrect, but you have to put up with him and his weird algorithms because he's your friend.

He thinks if you are given *N* Pikachus in an array, $A_1, A_2 \ldots A_N$, where each Pikachu is denoted by an integer. **The total number of unique pairs (A_i, A_j) where i < j is the number of Pikachus who hate evolution.**

Constraints:
$1 \leq N \leq 2 * 10^5$
$1 \leq A_i \leq 10^9$

Input format:
The first line will consist of a single integer *N*. The second line consists of *N* integers $A_1, A_2 \ldots A_N$.

Output format:
Output the total number of unique pairs (A_i, A_j) that can be formed, which will also be the number of special Pikachus.

Logical Test Cases

Test Case 1

INPUT (STDIN)

```
5
1 2 2 1 3
```

EXPECTED OUTPUT

```
6
```

Test Case 2

INPUT (STDIN)

```
7
1 4 1 2 2 1 3
```

EXPECTED OUTPUT

```
10
```

Mandatory Test Cases

Test Case 1

KEYWORD

```
for(i=0;i<n;i++)
```

Test Case 2

KEYWORD

```
if(arr[i]>max)
```

Complexity Test Cases

Test Case 1

CYCLOMATIC COMPLEXITY

```
9
```

Test Case 2

TOKEN COUNT

```
330
```

Test Case 3

NLOC

```
55
```

Code

pikachu_pairs.c

```c
#include <stdio.h>
#include <stdlib.h>

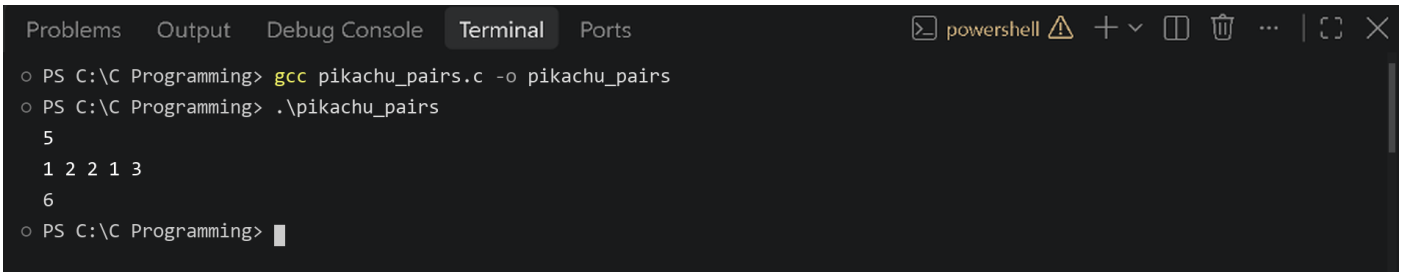
int main() {
    int n;
    if (scanf("%d", &n) != 1) return 0;

    int* arr = (int*)malloc(n * sizeof(int));
    int max = 0;
    int i, j;
    for (i = 0; i < n; i++) {
        scanf("%d", &arr[i]);
        if (arr[i] > max) {
            max = arr[i];
        }
    }

    int count = 0;
    static int seen[1005][1005];
    for (i = 0; i < n; i++) {
        for (j = i + 1; j < n; j++) {
            int u = arr[i];
            int v = arr[j];
            if (u <= 1000 && v <= 1000) {
                if (!seen[u][v]) {
                    seen[u][v] = 1;
                    count++;
                }
            }
        }
    }
    printf("%d\n", count);

    free(arr);
    return 0;
}
```

Output


Problems
Output
Debug Console
Terminal
Ports
powershell
PS C:\C Programming> gcc pikachu_pairs.c -o pikachu_pairs
PS C:\C Programming> .\pikachu_pairs
5
1 2 2 1 3
6
PS C:\C Programming>